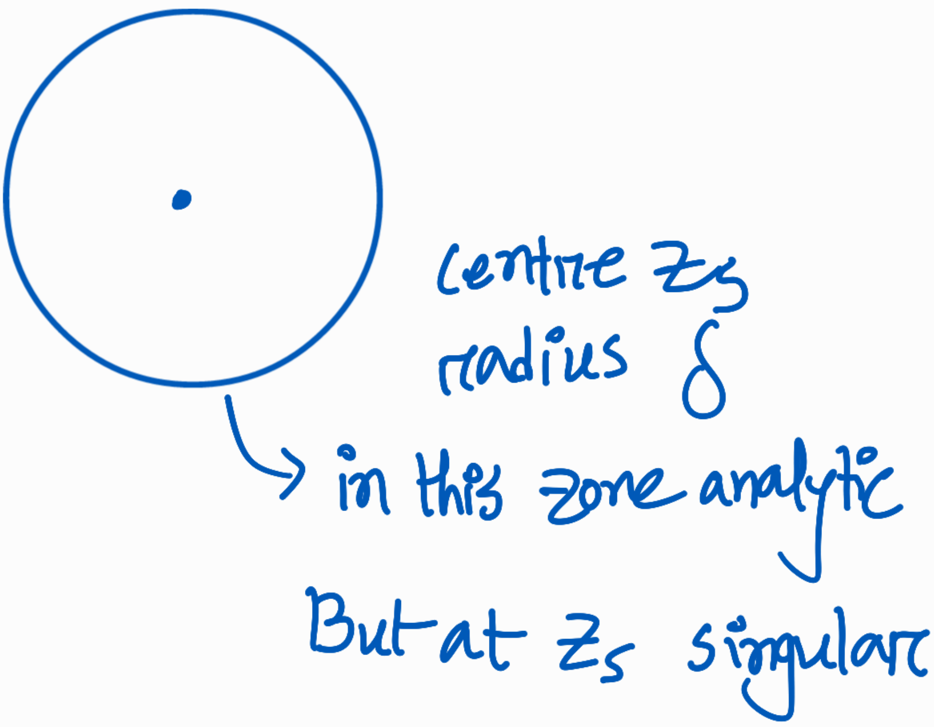
Singularity -

point at which function is not analytic. (not necessarily infinite)

Analytic function - If a complex func is differentiable at each point of its domain

Isolated Singularity

func singular at z_s but analytic for $0 < |z - z_s| < \delta$



Types -

- Removable singularity

if $w = f(z)$ infinite at $z = a$

$$g(z) = (z-a)^n f(z), \quad n \text{ integer}$$

if → n can be found to make $g(z)$ analytic at $z = a$

pole of $f(z)$ removed by forming $g(z)$

- Essential singularity

functions can have infinite num of terms which all approach infinity.

Singularity can not be removed by multiplying factors not poles. → These func. can be expanded in a descending power series of variable e.g. $e^{\frac{1}{z}}$, ess. singularity at $z=0$

$$w = e^{\frac{1}{z}} = 1 + \frac{1}{z} + \frac{1}{2!z^2} + \cdots + \frac{1}{n!z^n} + \cdots$$

infinite at origin

$$z^p \longrightarrow z^p w = z^p e^{\frac{1}{z}} = z^p + z^{p-1} + \frac{z^{p-2}}{2!} + \cdots + \frac{z^{p-n}}{n!} + \cdots$$

negative powers of z ✓

z approaches 0 and value tends to ∞

p can not be found.

Pole - singularity where func. val becomes infinite. Also called unessential singularity

$w = \frac{1}{z}$, $z=0$, infinite

So, w has pole at origin

pole has 'order'

$W = \frac{1}{z} \longrightarrow$ 1st order

$W = \frac{1}{z^2} \longrightarrow$ 2nd "

Non-Isolated Singularity

For multi valued functions

e.g. $f(z) = \dfrac{1}{\sin\left(\frac{1}{z}\right)}$

Taylor's Series

Every analytic function equals a convergent power series about any pt. in its domain

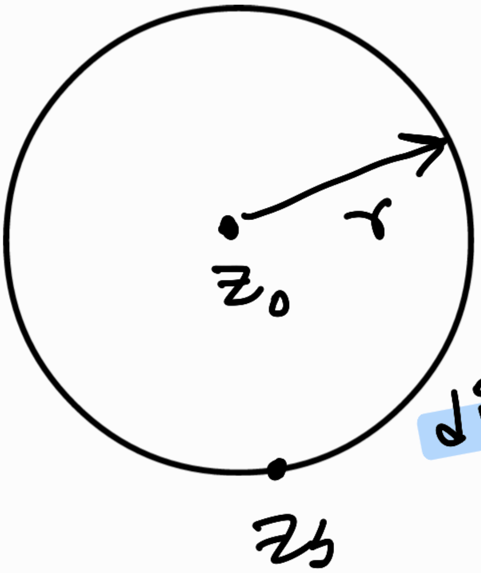
$f(z)$ analytic in region A

$z_0 \in A$

$f(z)$ can be expanded $\rightarrow$ (about z_0)

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

The series converges for all z such that $\underbrace{|z - z_0| < r}_{\text{entirely contained in } A}$



diverges outside

r depends on nearest z_s

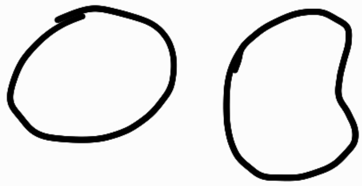
Coefficient for Taylor series

$$a_n = \frac{1}{2\pi i} \int_C \frac{f(z)}{(z-z_0)^{n+1}}\, dz$$

[From Cauchy's integral th.]

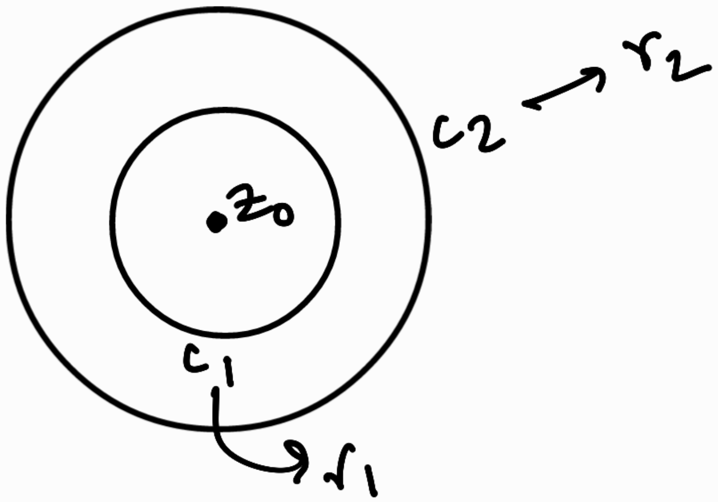
z is a point on curve C in complex plane

$$z(t) = x(t) + i y(t), \quad a \leq t \leq b$$

C is a closed curve  around z_0

C subset of A

Laurent's series



$f(z)$ analytic on the circles within annulus between c_1 and c_2

Then for any z in the annulus,

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n + \sum_{n=1}^{\infty} b^n (z - z_0)^{-n} \quad \text{.......} \quad (i)$$

(i) is the Laurent series of $f(z)$ about pt. $z = z_0$

$$a_n = \frac{1}{2\pi i} \int_{C_1} \frac{f(z)}{(z-z_0)^{n+1}} dz$$

$$b_n = \frac{1}{2\pi i} \int_{C_2} \frac{f(z)}{(z-z_0)^{-n+1}} dz$$

integration taken in positive direction

e.g. $z = \frac{\pi}{4}$,

$f(z) = \cos z$

$f(z) = \cos z \longrightarrow \frac{1}{\sqrt{2}}$

$f'(z) = -\sin z \longrightarrow -\frac{1}{\sqrt{2}}$

$f''(z) = -\cos z \longrightarrow -\frac{1}{\sqrt{2}}$

$$\cos z = \frac{1}{\sqrt{2}} + \left(z-\frac{\pi}{4}\right)\left(\frac{-\frac{1}{\sqrt{2}}}{1!}\right) + \left(z-\frac{\pi}{4}\right)\left(\frac{-\frac{1}{\sqrt{2}}}{2!}\right) + \cdots$$

Suppose, z_0 is isolated singular point of $f(z)$

Laurent expression about z_0 :

$$f(z) = \sum_{n=-\infty}^{\infty} c_n (z-z_0)^n, \quad 0<|z-z_0|<\delta$$

Residue :

Coefficient of $(z-z_0)^{-1}$ in the series

$$\text{Res}[f(z), z_0] = c_{-1}$$

$$= \frac{1}{2\pi i} \oint_C f(z)\,dz$$

In complex analysis, function represented as series – Laurent series

To integrate → term by term, but some integrals become 0.

What doesn't become 0 → Residue
↓
determines integral value

coefficient of $\frac{1}{z-a}$ → residue of $f(z)$ at $z=a$
↓
isolated singular point

- If $\lim_{z \to z_0} f(z)$ exists and finite,

↳ z_0 removable singularity

- If $\lim_{z \to z_0} f(z) = \infty$, pole

- $\lim_{z \to z_0} f(z)$ doesn't exist and $\neq \infty$, essential singularity

For removable singularity,

Residue $= 0$

Essential singularity,

Residue $= c_{-1}$ (may be non zero)

Pole

Using standard pole formulae

Laurent expansion of $f(z)$ about point where not analytic (a)

$$f(z) = \underbrace{a_0 + a_1(z-a) + a_2(z-a)^2 + \cdots}_{\text{analytic part}}$$

$$\underbrace{+ \frac{b_1}{(z-a)} + \frac{b_2}{(z-a)^2} + \cdots}_{\text{principal part}}$$

principal part → if 0, series becomes Taylor series

$$a_n = \frac{1}{2\pi i} \oint_{c_1} \frac{f(z)}{(z-a)^{n+1}}\, dz$$

$$b_n = \frac{1}{2\pi i} \oint_{c_2} \frac{f(z)}{(z-a)^{-n+1}}\, dz$$

Laurent Theorem → if we want to expand $f(z)$ about pt. a where $f(z)$ is not analytic, Laurent's series is used.

For complex number,

at simple pole, $\operatorname{Res} f(z) = \lim_{z \to a} (z-a) f(z)$

pole of order n,

$$\frac{1}{n-1}\left[\frac{d^{n-1}}{dz^{n-1}}\left\{(z-a)^n f(z)\right\}\right]_{z=a}$$

$$\hookrightarrow \lim_{z \to a} \frac{1}{(k-1)!} \frac{d^{k-1}}{dz^{k-1}}\left\{(z-a)^k f(z)\right\}$$

order k

Residue theorem

singularities $a, b, c \cdots$

Residues a_{-1}, b_{-1}, c_{-1}

$$\oint_C f(z)\,dz = 2\pi i \left(\underbrace{a_{-1} + b_{-1} + c_{-1} + \cdots}_{\substack{\text{sum of} \\ \text{residues of} \\ f(z) \\ \text{enclosed by} \\ C}} \right)$$